

FastAPI Request Timeout SOP

Objective

Resolve request timeout incidents caused by long-running operations, slow dependencies, worker exhaustion, or gateway timeout settings.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T022
Issue	FastAPI timeout issue
Category	Performance
Service	FastAPI
Severity	Medium
Status	Resolved
Description	Request timeout during processing
Expected Resolution	Optimize long-running operations or use async tasks

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L007	5/1/2026 10:03	ERROR	Request timeout occurred	API_TIMEOUT
L028	5/1/2026 10:14	WARNING	Request timeout occurred	API_TIMEOUT
L048	5/1/2026 10:24	WARNING	Request timeout occurred	API_TIMEOUT

Procedure

1. Confirm timeout status code, timeout duration, endpoint, and request payload.
2. Review logs for timeout messages and correlate with worker saturation or long task execution.
3. Check database, cache, and downstream API response times during the incident window.
4. Identify synchronous blocking calls inside async endpoints.
5. Validate API gateway, ingress, Uvicorn, and Gunicorn timeout settings.
6. Move long-running work to background tasks or queue-based processing when appropriate.
7. Optimize slow logic and set safe client and server timeouts.
8. Monitor timeout count, latency percentiles, and worker utilization after remediation.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Request timeout	Long-running operation	Use async task or optimize logic
Timeout under load	Worker exhaustion	Scale workers or replicas
Gateway timeout	Ingress timeout too low	Align gateway and app timeout settings
Timeout from dependency	Slow DB or external API	Tune dependency timeout and retry policy

Best Practices

1. Avoid heavy work in request-response paths.
2. Use timeout and retry policies for external calls.
3. Track timeout rate by endpoint.
4. Document expected SLA per API route.

References

FastAPI Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | API Gateway Logs | Backend Runbook